

Predictive and Generative AI

Project: Verified Study Buddy

1. Your mission

Your learning target

I can distinguish discriminative prediction from generation by inspecting the requested output.

I can calculate an empirical next-token distribution from counted continuations.

I can separate next-token fit from factual support and downstream usefulness.

Remember before reading

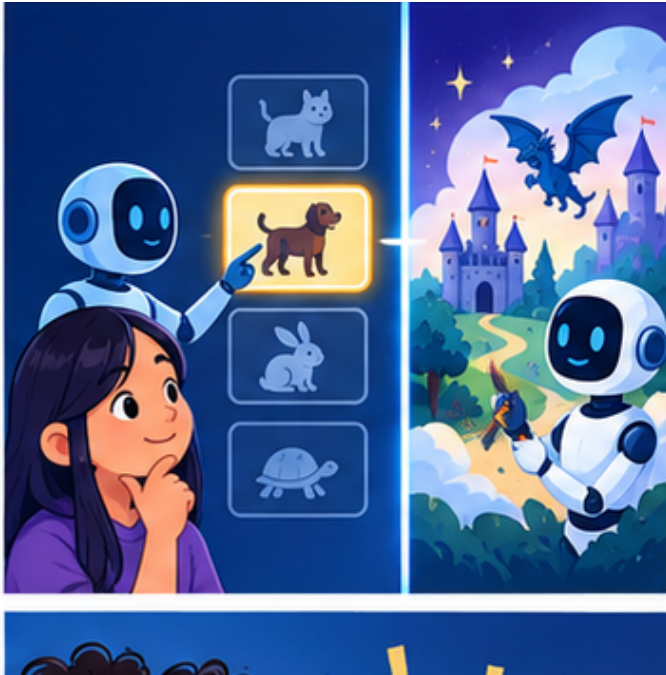
Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Capstone Expo. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 1: Predictive and Generative AI

Prediction estimates a target; generation samples a new structured output.



CREATOR MISSION

Classify systems and explain the output space for each.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

What is the difference between a model prediction and evidence that a statement is true?

This week's end goal

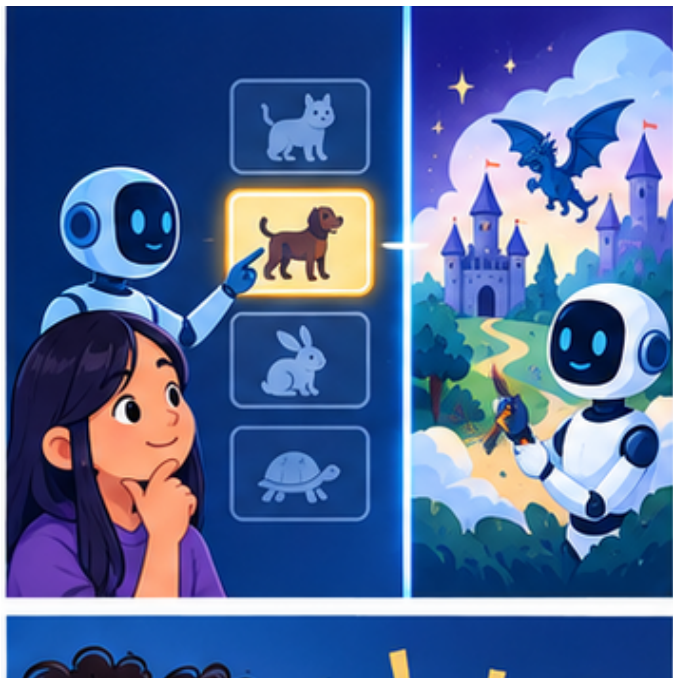
By the end, I can identify generation and prediction objectives without confusing either with verified truth.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use predictive and generative ai to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A classifier might map a message to the label request or report. A text generator instead produces a sequence, predicting a distribution over the next token given a prefix. Both make predictions, but the training targets and permitted outputs differ. One application can combine generation with a classifier, retrieval and human approval. Calling a whole product “generative” does not describe every component.

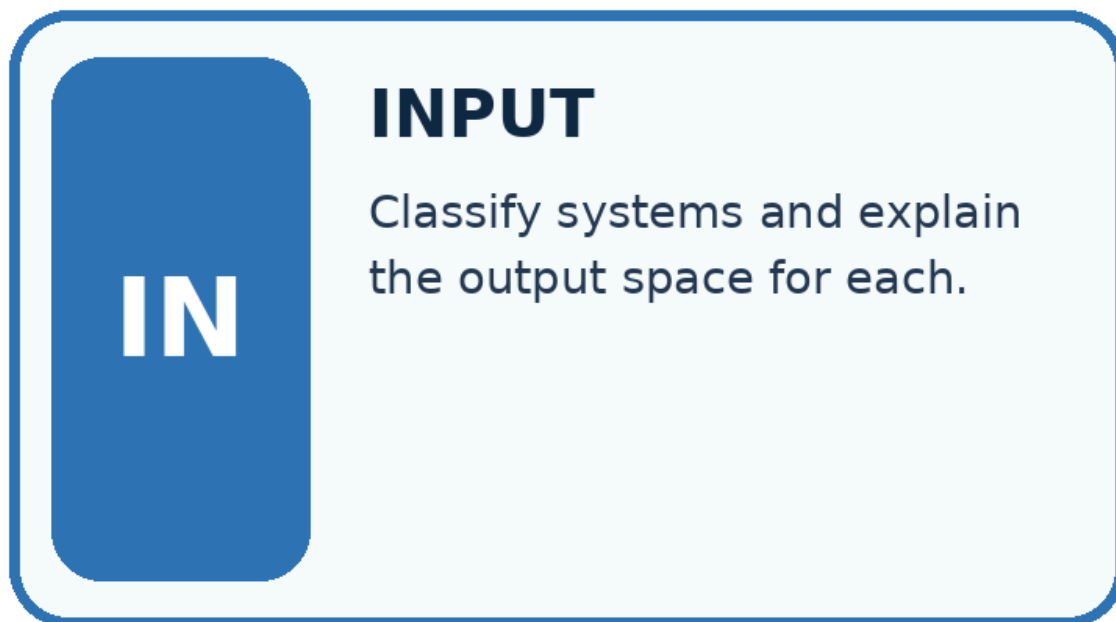
The objective defines what training attempts to fit. Next-token fit rewards matching observed continuations; it does not directly certify that an answer is true, well sourced or permitted. A fluent completion is still an output to evaluate against the task.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



SYSTEM

Prediction estimates a target; generation samples a new structured output.





EVIDENCE

Classify systems and explain the output space for each.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Prefix · Continuation · Count

Prefix: The marker is

Continuation: red

Count: 6

Prefix: The marker is

Continuation: blue

Count: 3

Prefix: The marker is

Continuation: green

Count: 1

There are ten observed continuations. The empirical probabilities are red $6/10 = 0.6$, blue $3/10 = 0.3$, green $1/10 = 0.1$. They are nonnegative and sum to 1. A simple frequentist classroom model uses these relative counts as its next-token distribution. These probabilities describe the supplied corpus counts, not the current colour of any actual marker.

Keep the evidence roles separate

Plan or input · Procedure · Permitted conclusion

Plan or input: Ten corpus continuations

Procedure: Count red 6, blue 3, green 1

Permitted conclusion: Distribution 0.6, 0.3, 0.1

Plan or input: Prefix: The marker is

Procedure: Select a next token

Permitted conclusion: An output, not an observation

Plan or input: Current marker question

Procedure: Check direct evidence

Permitted conclusion: Colour claim needs support

The continuation model and a factual observation have different evidence sources.

Connect the picture and the explanation

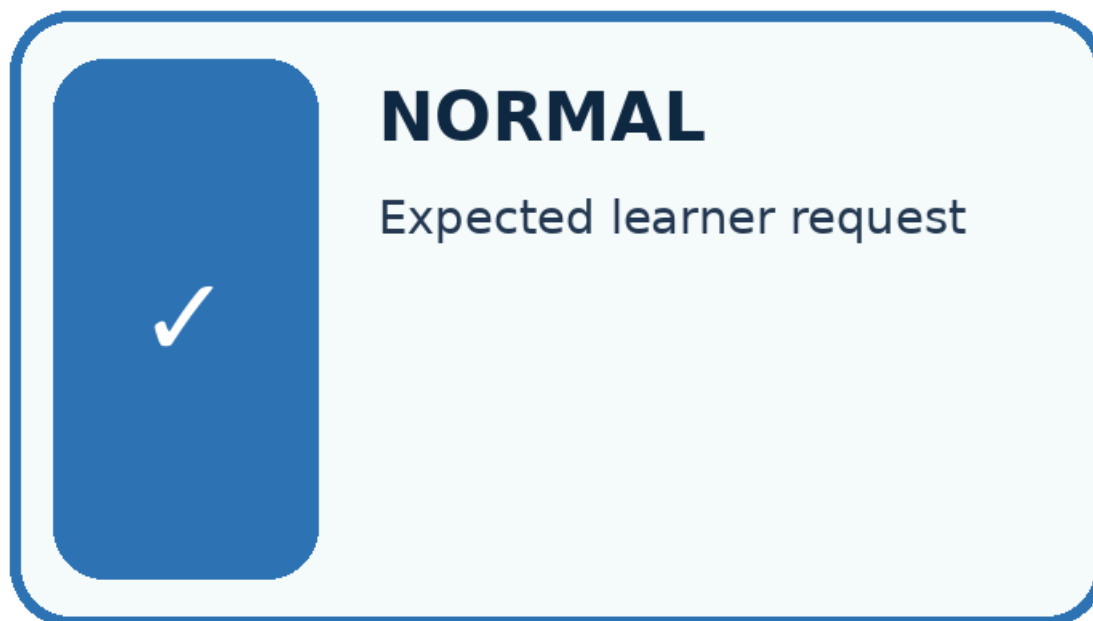
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.





AMBIGUOUS

Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING SOURCE

Evidence unavailable



MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: The classifier instead predicts history while N and the draft stay unchanged.

Prediction: The category prediction changes, but the same unsupported draft claim still needs checking.

Reason: Stages can fail independently; one result does not validate another.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

PREDICTIVE AI

estimates a label, score, or number

2

GENERATIVE AI

constructs new content from learned patterns

3

TARGET

value a predictive system estimates

4

OUTPUT SPACE

set of possible results

Word · My own example · Not this / boundary

Word: predictive AI

My own example: AI used to estimate a target such as a category, number or future outcome from inputs.

Not this / boundary: _____

Word: generative AI

My own example: AI that produces new content such as text, images or audio rather than only selecting a

category.

Not this / boundary: _____

Word: target

My own example: The category, value or outcome a prediction task is intended to estimate.

Not this / boundary: _____

Word: output space

My own example: The set or range of outputs a system can produce for a task.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Generative objective

Meaning: A training aim concerned with producing or modelling possible outputs

Word: Discriminative objective

Meaning: A training aim mapping inputs to a specified target such as a label

Word: Conditional distribution

Meaning: Probabilities for alternatives given stated information

Word: Autoregressive generation

Meaning: Producing a sequence one next item at a time using preceding items

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 1 • WORKED EVIDENCE

- 1 Classifier: choose one label from cat, dog, or rabbit.
- 2 Regressor: estimate a numerical reading time.
- 3 Generator: construct a new explanation token by token.
- 4 A product may combine mechanisms, so test each stage separately.

1. Classifier: choose one label from cat, dog, or rabbit.
2. Regressor: estimate a numerical reading time.
3. Generator: construct a new explanation token by token.
4. A product may combine mechanisms, so test each stage separately.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

After producing red, the next step conditions on the expanded prefix “The marker is red”. A second table might give an end token probability 0.8 and the token today probability 0.2. A generated sequence can stop at the end token or continue. Do not reuse the first-step table unchanged after adding a token unless the model explicitly ignores the new prefix.

A separate classifier can return red as a label from an image; that is a different input and target from continuing text. A downstream evaluation asks whether the system correctly reports the current marker. It must include ground truth or sources and appropriate failure handling, not just count plausible phrases.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Predictive and Generative AI

A fictional learning tool first classifies a question, then drafts an answer from a supplied note.

The facts we will use

A fitted classifier labels a question science; its training used labelled questions.

Approved note N: evaporation changes liquid water into water vapour. A generator drafts: “Water evaporates into vapour and always disappears forever.”

Predictive and Generative AI

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Predict category

The classifier predicts science from the new question.



2. Retrieve note

N supplies a specific statement about evaporation.



3. Generate draft

The draft adds "always disappears forever" beyond N.



4. Verify claims

Keep the supported change-of-state claim; remove or review the unsupported addition.

CHANGE ONE THING

The classifier instead predicts history while N and the draft stay unchanged.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Predict category

What happens and why: The classifier predicts science from the new question.

Step: 2. Retrieve note

What happens and why: N supplies a specific statement about evaporation.

Step: 3. Generate draft

What happens and why: The draft adds “always disappears forever” beyond N.

Step: 4. Verify claims

What happens and why: Keep the supported change-of-state claim; remove or review the unsupported addition.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Classification predicts a category under a trained mapping. Generation produces a sequence of words under a language objective. Supplying a note can support factual grounding, but the draft still requires claim-level checking. The category being correct does not prove every sentence in the answer is correct.

What this example does not establish

This example does not expose a real model’s architecture or prove that source-fed generation always stays grounded.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: Every AI system that produces an answer is generative. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Keep three questions separate: did the model fit continuation data; did the output satisfy the requested format or task; and are factual claims supported? A system that drafts a fictional description has different

criteria from one reporting a live measurement. Record the source and time when a claim refers to changing conditions. No AI account is needed to reason about these distinctions.

Exit: a team says “the highest-probability continuation proves the current fact”. Name the evidence they actually have and the evidence still needed.

8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Supported case: the prefix “The door is” is followed in a tiny corpus by open four times and closed once. The empirical probabilities are $4/5$ and $1/5$. A current sensor reports closed. Selecting the most frequent continuation open would be plausible under the corpus model but inconsistent with the sensor observation.

Guided: compute both probabilities and explain why selecting open does not establish the current door state.

Practise with less help: Compute the distribution

Use workbook W2 for the connected example “Generative and predictive objectives”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Generative and predictive objectives. Proposed explanation: “Generative models do not predict”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Identify the first unsupported claim and explain what source would be needed before asserting it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → RUN → VERIFY → EXPLAIN

Sort twelve systems by mechanism. For each, name its output space, evidence needed, and one possible failure.

Sort twelve systems by mechanism. For each, name its output space, evidence needed, and one possible failure.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY
1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



VERIFIED STUDY BUDDY

Separate Study Buddy tasks into classify, calculate, retrieve, and generate stages.

THIS WEEK'S CHECKPOINT Separate Study Buddy tasks into classify, calculate, retrieve, and generate stages.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

Evidence for your project

For the Verified Study Buddy, separate routing predictions, source retrieval and generated explanations in its evidence record.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT CREATOR PROOF

Classify systems and explain the output space for each.

CHALLENGE Classify systems and explain the output space for each.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 1 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A tool labels a question biology, writes a new question and computes 7 times 8.

The question

Identify each operation.

Worked explanation

Labelling selects a category; generation creates text; calculation returns 56. Generated questions need factual and instructional checks.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Three outputs need three different checks

Evidence or stage · Value or result

Evidence or stage: Classify

Value or result: Select biology as a category

Evidence or stage: Generate

Value or result: Compose a new question

Evidence or stage: Calculate

Value or result: 7 times 8 equals 56

A tool labels a question biology, writes a new question and computes 7 times 8.

Identify each operation.

Labelling selects a category; generation creates text; calculation returns 56. Generated questions need factual and instructional checks.

14. Try a new case independently

Independent case: Changed independent case

A system retrieves an approved Moon note, uses a trained classifier to predict reading difficulty, then writes a poem. Identify each mechanism and the evidence needed to check its output.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 1 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A system labels a message as spam. Another writes a new reply. Which task predicts a category and which generates content? Can the new reply contain a false fact?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Separate Study Buddy tasks into classify, calculate, retrieve, and generate stages.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

A study tool finds a paragraph, labels its difficulty and invents practice questions. Identify the three operations and one check for each.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.